

TILLABERI, Niger

WMO: 610360

Lat: 14.200N

Lon: 1.450E

Elev: 689

StdP: 14.33

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	61.8	64.0	17.4	13.6	85.1	18.3	14.1	84.4	19.9	78.3	17.2	77.9	1.4	40	0.407

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	21.3	109.7	71.6	108.0	71.1	106.5	71.1	80.0	93.6	79.3	93.0	78.8	92.6	4.7	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.2	145.3	85.1	76.3	141.0	84.7	75.6	137.4	84.4	44.2	93.5	43.4	92.6	42.8	92.3	92.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.9	12.7	11.0	DB	56.2	112.8	2.9	2.1	54.1	114.3	52.4	115.5	50.8	116.7	48.7	118.2
			WB	44.0	84.8	1.5	3.1	42.9	87.0	42.1	88.8	41.3	90.6	40.2	92.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	86.9	77.4	82.7	88.9	94.3	95.5	91.7	86.7	84.5	87.0	89.8	85.3	79.2
	DBStd	6.50	3.93	4.95	4.10	3.30	3.57	4.17	3.99	3.23	3.53	2.62	3.05	3.57
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	13475	850	915	1205	1330	1410	1251	1137	1068	1110	1233	1060	906
	CDD65	8000	385	495	740	880	945	801	672	603	660	768	610	441
	CDH74	114844	3980	6301	11262	14726	15788	12600	9221	7498	9015	11377	8257	4819
	CDH80	71379	1876	3626	7368	10517	11357	8381	5108	3582	4980	7131	5003	2450
Wind	WSAvg	3.7	4.2	4.6	4.5	3.6	4.5	4.9	4.0	3.1	2.6	2.3	2.6	3.4
Precipitation	PrecAvg	15.7	0.0	0.0	0.1	0.1	0.7	2.1	4.1	5.9	3.1	0.4	0.0	0.0
	PrecMax	27.5	0.0	0.4	1.4	0.9	3.4	5.0	9.1	14.1	14.5	5.2	0.4	0.2
	PrecMin	8.3	0.0	0.0	0.0	0.0	0.0	0.3	1.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	4.1	0.0	0.0	0.3	0.2	0.8	1.1	1.9	2.8	2.6	0.8	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.8	105.4	108.2	112.1	111.9	108.1	103.0	98.2	102.9	105.2	103.1	99.1
		MCWB	62.5	66.6	66.5	70.8	72.3	75.1	76.5	78.4	76.9	72.0	67.3	64.4
	2%	DB	95.0	101.9	106.4	110.0	109.8	105.9	100.6	96.2	100.2	103.4	101.6	96.3
		MCWB	61.2	65.1	66.2	70.3	73.2	75.8	76.9	78.0	77.4	72.4	67.0	63.0
	5%	DB	92.3	98.9	104.6	108.4	108.1	103.9	98.4	94.3	97.9	101.8	100.0	94.1
		MCWB	60.2	63.6	65.7	69.7	73.5	76.0	76.7	77.9	77.5	72.5	66.3	61.9
	10%	DB	89.5	96.0	102.6	106.8	106.3	101.7	96.0	92.5	95.8	100.2	98.2	91.8
		MCWB	59.3	62.1	65.0	69.4	74.0	75.9	76.7	77.6	77.5	72.7	65.5	61.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.1	71.2	75.4	78.3	79.9	79.8	80.1	80.8	81.1	80.4	73.9	66.6
		MCDB	93.5	93.6	89.5	101.7	101.5	96.2	92.3	92.3	93.6	95.7	93.2	94.2
	2%	WB	63.0	67.9	71.5	76.4	78.7	78.8	79.1	79.8	80.0	78.9	71.3	65.0
		MCDB	90.8	97.4	93.6	99.3	99.2	96.5	91.9	91.1	92.6	94.8	92.5	92.4
	5%	WB	61.4	65.4	68.7	75.3	77.9	78.1	78.3	79.0	79.2	78.0	69.2	63.5
		MCDB	89.1	94.2	99.3	98.4	98.5	95.7	91.2	90.0	91.9	93.8	92.1	90.4
	10%	WB	60.0	63.6	66.7	74.1	77.1	77.4	77.8	78.4	78.7	77.1	67.5	62.0
		MCDB	87.1	92.7	98.4	97.4	97.8	95.0	90.6	88.8	91.2	92.9	91.9	88.2
Mean Daily Temperature Range	5% DB	MDBR	24.7	25.7	26.4	25.7	21.3	18.6	16.2	14.9	17.1	22.2	27.1	25.4
		MCDBR	28.4	29.4	29.2	27.8	23.8	20.7	19.0	17.3	19.9	24.3	28.2	28.3
		MCWBR	10.2	9.7	9.0	8.5	6.4	4.4	4.1	4.5	4.7	6.5	8.1	9.4
	5% WB	MCDDBR	26.8	27.7	25.7	22.4	20.2	18.8	16.7	15.9	17.4	20.0	26.5	28.3
		MCWBR	10.6	9.9	9.6	7.8	6.0	5.1	5.0	5.5	5.4	5.8	8.8	9.9
Clear-Sky Solar Irradiance	taub	0.583	0.622	0.712	0.737	0.743	0.724	0.660	0.613	0.596	0.635	0.559	0.542	
	taud	1.717	1.595	1.442	1.484	1.483	1.535	1.698	1.784	1.805	1.682	1.735	1.774	
	Ebn at Noon	227	226	210	206	201	202	216	229	234	220	232	234	
	Edh at Noon	73	85	102	97	96	90	77	71	70	77	71	67	
All-Sky Solar Radiation	RadAvg	1758	1925	2036	2084	2056	1934	1890	1842	1963	1917	1809	1704	
	RadStd	71	87	102	94	56	83	41	70	52	58	42	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.67	+1.69	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page